Project Design Phase-II
Technology Stack (Architecture & Stack)

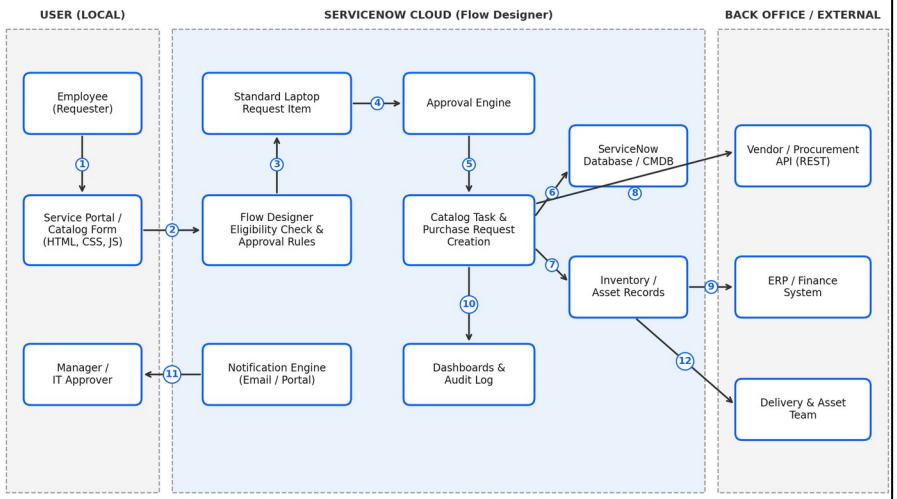
Date	29 September 2026
Team ID	SWTID-2026-7788
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Automated standard laptop order processing in ServiceNow

Reference: ServiceNow Flow Designer and Service Catalog documentation



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employee raises a standard laptop request through the Service Catalog form; requester tracks status in the portal	ServiceNow Service Portal (HTML, CSS, JavaScript, Angular JS widgets)
2.	Application Logic-1	Flow Designer flow: validates the model is standard and within budget, then auto-approves or routes for approval	ServiceNow Flow Designer, Decision / Approval actions
3.	Application Logic-2	Creates catalog task, purchase request and stock reservation automatically after approval	Flow Designer, Subflows, Server-side JavaScript
4.	Application Logic-3	Sends email / portal notifications on submission, approval, dispatch and delivery	ServiceNow Notifications, Email Scripts
5.	Database	Stores requests, approvals, catalog items and user / department data	ServiceNow platform database (relational)
6.	Cloud Database	Hosted database service on the ServiceNow cloud instance	ServiceNow Cloud (managed instance), CMDB
7.	File Storage	Stores quotes, approval attachments and delivery documents	ServiceNow Attachments (platform storage)
8.	External API-1	Sends purchase orders to the vendor / procurement system and receives order status	REST / SOAP API via IntegrationHub
9.	External API-2	Syncs cost center and budget checks with the finance / ERP system	REST API, IntegrationHub spoke
10.	Machine Learning Model	Not used in the current scope; optional future use for demand forecasting of laptop stock	N/A (optional: ServiceNow Predictive Intelligence)

11.	Infrastructure (Server / Cloud)	Application deployed on the ServiceNow cloud instance. Local Server Configuration: not required. Cloud Server Configuration: ServiceNow managed instance (Dev / Test / Prod)	ServiceNow Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source and standard web frameworks used in portal widgets and scripts	AngularJS, Bootstrap, JavaScript (Rhino)
2.	Security Implementations	Role-based access for requesters, approvers and procurement; approvals audited; data encrypted in transit and at rest	ACLs, IAM / SSO roles, TLS 1.2+, AES-256, OWASP guidelines
3.	Scalable Architecture	Multi-tier platform architecture; logic kept in reusable flows and subflows so new models / departments need only configuration	3-tier architecture, Flow Designer subflows
4.	Availability	Managed cloud instance with data-center redundancy, failover and backups	ServiceNow Cloud, load balancers, disaster recovery
5.	Performance	Automated approvals reduce manual steps; async flow execution and cached catalog data keep response fast	Asynchronous flows, platform caching, indexed tables

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d